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Submitted in response to: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")

The NSF-Simons AI Institute for Cosmic Origins (CosmicAI) supports the Office of Science and Technology's mission to deploy strategies that support and grow responsible AI use and development, which is critical to maintaining US competitiveness in scientific discovery and technological innovation.

Overview

Understanding the formation and evolution of the Universe and our place within it is one of the most foundational goals of human inquiry. The recent revolution in AI, advances in telescope capabilities, and accelerating computing capabilities, have opened new frontiers of scientific analysis, enabling astronomers to tackle complex, data-intensive questions related to our cosmic origins: What is the nature of dark matter and energy? What were the earliest chapters in our cosmic history? How do supermassive black holes at the center of galaxies like our Milky Way form and grow? Is there evidence for life outside our Solar System?

Federal support for AI development and application is imperative – in astronomy and beyond – to:

1. strengthen US leadership and competitiveness in technological innovation and AI-driven scientific discovery
2. advance fundamental and applied interdisciplinary research about our Universe
3. capitalize on our national investment in existing ground-based and space-based infrastructure and facilities
4. train the next generation of leaders in science, technology, engineering, and math (STEM)

NSF-Simons AI Institute for Cosmic Origins: Who we are

NSF-Simons AI Institute for Cosmic Origins (CosmicAI) is an astrophysics AI research institute jointly funded by the NSF and Simons Foundation as part of the NSF National Artificial Intelligence Research Institutes program. We aim to develop transformative AI methods to meet pressing astronomical challenges and tackle outstanding questions about our cosmic origins through research in four AI themes: trustworthiness, efficiency, interpretability, and robustness. CosmicAI also aims to serve as a nexus of collaboration to increase access to astronomy and AI data and methods through open-source, AI-powered tools, data sharing, and AI educational initiatives.

Any opinions or recommendations expressed in this statement are those of the authors and do not necessarily reflect the views of the National Science Foundation or the Simons Foundation.

AI for Science

In the context of astrophysics, AI-driven research goes far beyond large language models, such as ChatGPT, which have recently captured the public imagination. **Within science, AI techniques are used to conduct intelligent analyses of scientific data, improve the modeling of physical systems, and discover patterns within data.** The application of AI within scientific domains aims to create intelligent algorithms that learn from scientific data without being explicitly programmed. As part of responsible AI, the field of eXplainable Artificial Intelligence (XAI) has a special focus on science: understanding the *why* behind AI predictions. Explainability is crucial to build trust in AI applications and to promote responsible adoption and use.

Why AI + Astronomy

The field of astronomy has unique characteristics that serve as an ideal testbed for AI research:

- Astronomical data, unlike other scientific data, are public, plentiful, and non-proprietary;
- Astronomy research occurs in an open-source, collaborative ecosystem that provides unparalleled opportunities to develop and test AI methods in a “safe” public domain;
- Astronomical data are complex (multi-modal), noisy and high-dimensional, presenting a challenging application space that offers extensions to a broad range of other fields.

The growth of AI is powered by large amounts of rich data, which are necessary to expedite the continuing advancement and tangible outcomes of its use. Astronomy has a long history in large, well-understood data driving discovery and yielding quantitative, reproducible outcomes: verification of the law of gravity and Einstein's Theory of Relativity, the discovery of dark matter, and the expansion of the Universe. As an area of fundamental research, Astronomy is well positioned to assist AI researchers to safely experiment and develop new capabilities without being burdened by the legal and financial restrictions often associated with other data. AI research with astronomical data will foster new breakthrough methods of discovery vetted by extensive data.

Astronomy, through the wonder of the night sky, is a gateway to STEM fields. When combined with the development and extension of AI technologies, the universal appeal of astronomy is a springboard that can accelerate the training of our future AI workforce. Astronomical data is both rich and open-source, so students of all ages can engage with AI through interaction with inspiring data to grow areas of computer vision, robotics, remote sensing, and data-driven modeling of physical processes. Experience gained in an astronomical data learning environment will continue expanding our leadership in foundational research and industrial applications.

(1) Maintaining Competitiveness in Scientific Discoveries and AI Advancement

Maintaining the U.S. global dominance in science and technology requires dedicated federal investment in AI-driven research and training.

Maintaining our U.S. advantage in scientific research requires a collective impact approach in which interdisciplinary partners – industry, government, non-profits, and academic institutions – commit complementary knowledge and resources. New AI innovations across sectors drive the development of new technologies and transformational discoveries; AI is now evolving hand-in-hand with scientific progress. In the context of astronomy research, and scientific research more broadly, private sector investments in AI do not adequately support the needed foundational research that can bring about leaps in progress. Without continued federal investment in fundamental research and educational programs, we will lose our global competitiveness

(2) Advancing AI-Driven Discoveries in Astronomy

Foundational breakthroughs are seeded by foundational scientific research.

The use of AI in astrophysics research and related areas has grown significantly over the last decade. We are now at the crossroads of a genuinely transformative paradigm shift in which AI plays a central role as a tool of discovery. AI is being deployed in cutting-edge astronomy research across the following areas:

- **Dynamic and High-Throughput Data Analysis:** Big-data processing and signal detection (“anomaly detection”) expedite the automated processing of telescope observations and enable the rapid analysis of complex data streams from electromagnetic, astroparticle, and gravitational wave observatories, i.e., multi-messenger astrophysical observations.
- **Accelerating Models (“emulators”) for Physical Systems:** AI modeling methods offer potential speedups of hundreds to thousands of times over classic numerical approaches, enabling transformative progress in the current size and resolution limits of computational models of the Universe.
- **Creating Transparency in AI Decisions:** XAI is a growing research area that aims to make AI reasoning more transparent. In order for AI models and predictions about the Universe, and science more broadly, to be trustworthy, AI decisions and analyses must be interpretable.
- **Exploration and Synthesis of Text and Numerical Data:** Extensions to models, such as ChatGPT, offer the ability to create a radically new environment for conducting and disseminating research. Such models can answer queries about astronomy knowledge, accomplish complex coding tasks rapidly, and generate scientific data (in lieu of direct numerical solution).

AI-driven progress in the above areas will advance fundamental research into the nature of our Universe, while seeding advancements in other disciplines, ranging from combustion and weather modeling to robotics.

(3) Optimizing our Investment in Existing Astronomical/Space Infrastructure

AI is an essential tool to maximize the scientific return from and financial investment in U.S. telescope facilities.

We are at the forefront of an era where astrophysical research is increasingly driven by Big Data, fundamentally reshaping how discoveries are made both theoretically and observationally.

The 2020 National Academies Decadal Survey of Astronomy & Astrophysics¹ identified AI as a critical component for analyzing data from our next-generation astronomical observatories. Nearly every subfield of astronomy is being transformed into a data-intensive science spanning research areas such as galaxy formation and evolution, time-domain studies that track variation in the night sky caused by supernovae and other energetic events, and multi-messenger astronomy in which data are composites of observations from multiple ground-based and space-based telescopes. With datasets growing exponentially, AI is now an essential tool to reliably maximize the scientific return from the vast and complex information from our nation's telescope facilities, including the Very Large Array, Zwicky Transient Facility, Rubin Observatory, and James Webb Space Telescope.² Machine learning and neural networks enable researchers to process, classify, and analyze massive datasets with unprecedented speed and precision, uncovering new insights that would otherwise be lost in the overwhelming volume of data. This shift is especially critical as future facilities, which are essential to ensure that the U.S. remains a global leader in basic research, start to come online and generate even larger data sets. AI is required to maximize their scientific return on investment, ensuring that researchers can process and analyze data at unprecedented scales.

Beyond powering future discoveries, AI is also revolutionizing how we explore over half a century of archived astronomical data. These historical observations hold untapped potential, containing valuable information that earlier analysis techniques may have overlooked. AI can reprocess these data, revealing previously undetected astrophysical phenomena while enabling cross-correlation of observations across different wavelengths and missions. By harnessing AI to unify past and present data, astronomers can construct a more comprehensive picture of the universe, ensuring that every bit of information contributes to groundbreaking scientific advancements while reinforcing the U.S.'s leadership in the field.

(4) Training the Next Generation of Leaders in Science, Technology, Engineering, and Math

¹ The findings and recommendations of the 2020 Astronomy & Astrophysics decadal survey, which is a community-driven, holistic review of the entire field, can be found here:
<https://www.nationalacademies.org/our-work/decadal-survey-on-astronomy-and-astrophysics-2020-astro2020>

² <https://public.nrao.edu/telescopes/vla/>
<https://www.ztf.caltech.edu/>
<https://rubinobservatory.org/>
<https://webbtelescope.org/home>

A well-trained, technologically-skilled national workforce is essential to AI advancement and maintaining global competitiveness.

Recent job market data highlights a substantial and sustained increase in demand for advanced AI expertise across various sectors. According to LinkedIn's 2025 Emerging Jobs Report, AI Engineer and AI consultant are the two fastest growing jobs, reflecting a broadening adoption of AI-driven solutions in industry.³ A 2023 Stanford University report found that AI-related job postings grew at a remarkable rate of 74% annually between 2017 and 2022.⁴ Meanwhile, AI and data skillsets have the fastest growing demand, with 39% of current skills expected to be outdated by 2030.⁵ These trends underscore the expanding role of AI in shaping the future workforce. This growth is driven by increased investments in AI technologies across healthcare, finance, and manufacturing. With AI reshaping both industry and academia, the need for advanced, scalable education programs that equip professionals with cutting-edge AI skills is more critical than ever.

A strong national workforce is essential to bring new perspectives and approaches to AI research and drive the development of new ideas. Building a “deep bench” of experts in computing, engineering, science, and math ensures that the U.S. maintains its world leadership in scientific excellence, technological innovation, and global competitiveness in AI. Success requires innovative, integrated solutions for AI training and development that support and strengthen teachers at U.S. colleges and universities.

Despite the rapid growth of AI in scientific research, both in the public and private sectors, formal advanced training in AI has not kept pace with the rising demand for expertise. Astrophysics research, which engages a broad range of undergraduate students, graduate students, and post-PhD researchers, provides intensive training in wrangling, modeling, and predicting on large, complex, noisy datasets. Consequently, research and training in academic settings provides a crucial pipeline of personnel into the private sector, since more than half of these junior researchers ultimately transition to industry jobs.⁶

A dedicated investment that effectively builds the national workforce to support AI advances, and development requires:

³ “LinkedIn Jobs on the Rise 2025: 25 U.S. Roles That Are Growing in Demand | LinkedIn”
<https://www.linkedin.com/pulse/linkedin-jobs-rise-2023-25-us-roles-growing-demand-linkedin-news/>.

⁴ “AI Index Report 2024 – Artificial Intelligence Index,”
<https://aiindex.stanford.edu/report/>.

⁵ “The Future of Jobs Report 2025,” World Economic Forum,
<https://www.weforum.org/publications/the-future-of-jobs-report-2025/>.

⁶ “New Astronomy PhDs: What Comes Next - AIP.ORG,” AIP, January 25, 2023,
<https://ww2.aip.org/statistics/new-astronomy-phds-what-comes-next>.

- **Funding the training of students through foundational and applied science research.**
- **Supporting the educational mission of colleges and universities.**

Summary Recommendations

- Dedicate significant federal investment in AI-driven fundamental research programs sponsored by the NSF and other national agencies such as NASA and DOE.
- Invest in AI development and AI-use research that will maximize the scientific return from and financial investment in U.S. telescope facilities.
- Support U.S. universities and colleges in their mission to educate and train the next generation workforce of AI-skilled scientists, technicians, and researchers.
- Promote and support the development of *open-source* AI tools, training sets and models, which will elevate responsible AI use, promote trustworthy outcomes, and facilitate national adoption.

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